

Cyberdeck Cellular & Media — Cheat Sheet

Last-minute recall reference. For the reasoning behind each row, see [LESSON.pdf](#) or the interactive tour.

TASK	COMMAND / RULE
Check the modem is alive	AT → OK
Check signal quality	AT+CSQ
Check registration	AT+CREG? — 2nd field: 1=home, 5=roaming, 0=not registered
Check current carrier	AT+COPS?
Set/check APN	AT+CGDCONT=1,"IP","<apn>" / AT+CGDCONT?
SMS text mode (not PDU)	AT+CMGF=1 — set before AT+CMGS
Send SMS	AT+CMGS="+1..." then body, end with Ctrl+Z
Dial a voice call	ATD+1...; — trailing semicolon = voice, not data
Answer / hang up	ATA / AT+CHUP
Registered but no data?	Check the APN first, not the antenna
Not registered at all?	Check band/antenna/SIM, not the APN
Antennas needed	Both main + diversity — one alone still "works" but degraded
CSQ to dBm	$\text{dBm} = -113 + 2 \times \text{CSQ}$ (CSQ 0-31; 99 = unknown)
Ground-loop hum	Electrical problem — not fixable in ALSA/PulseAudio config
Power sizing	Size for the transmit-burst peak (1.5-2A), not the idle average (~200mA)